

6.1.4 - Einstein Eq.ⁿ

Recall that our matter evolution equations depend on Φ and Ψ (metric potentials).

$\Rightarrow$ we require evolution equations for metric perturbations $\leadsto$ Einstein equation

Let us first work to perturb $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$

$\uparrow$
 Ricci
 tensor
 (1)

$\uparrow$
 Ricci
 scalar
 (2)

(1) Recall that we can write $R_{\mu\nu} = \partial_\lambda \Gamma_{\mu\nu}^\lambda - \partial_\nu \Gamma_{\mu\lambda}^\lambda + \Gamma_{\lambda\rho}^\lambda \Gamma_{\mu\nu}^\rho - \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\rho}^\lambda$

Using our previous results,

$$R_{00} = -3\mathcal{H}' + \nabla^2\Psi + 3\mathcal{H}(\Phi' + \Psi') + 3\Phi''$$

$$R_{0i} = 2\partial_i(\Phi' + \mathcal{H}\Psi)$$

$$R_{ij} = [\mathcal{H}' + 2\mathcal{H}^2 - \Phi'' + \nabla^2\Phi - 2(\mathcal{H}' + 2\mathcal{H}^2)(\Phi + \Psi) - \mathcal{H}\Psi' - 5\mathcal{H}\Phi']\delta_{ij} + \partial_i\partial_j(\Phi - \Psi)$$

Note: for the derivations, see handwritten notes. (6 pages!!)

(2) We compute the Ricci scalar as follows:

$$R = g^{00}R_{00} + 2g^{0i}R_{0i} + g^{ij}R_{ij}$$

(1)
(2)
(3)

(1) $g^{00}R_{00} = -a^{-2}(1+2\Psi)[-3\mathcal{H}' + \nabla^2\Psi + 3\mathcal{H}(\Phi' + \Psi') + 3\Phi'']$

(2) $2g^{0i}R_{0i} = 0$ in Newtonian gauge, but more generally $\sim O(2)$

(3) $g^{ij}R_{ij} = -a^{-2}(1+2\Phi)\delta^{ij}[\mathcal{H}' + 2\mathcal{H}^2 - \Phi'' + \nabla^2\Phi - 2(\mathcal{H}' + 2\mathcal{H}^2)(\Phi + \Psi) - \mathcal{H}\Psi' - 5\mathcal{H}\Phi']\delta_{ij} + \partial_i\partial_j(\Phi - \Psi)$
 $= -a^{-2}(1+2\Phi)(3[\dots] + \nabla^2(\Phi - \Psi))$

$$\Rightarrow a^2 R = (1-2\Psi)[3\mathcal{H}' - \nabla^2\Psi - 3\mathcal{H}(\Phi' + \Psi') - 3\Phi''] + 3(1+2\Phi)[\mathcal{H}' + 2\mathcal{H}^2 - \Phi'' + \nabla^2\Phi - 2(\mathcal{H}' + 2\mathcal{H}^2)(\Phi + \Psi) - \mathcal{H}\Psi' - 5\mathcal{H}\Phi'] + (1+2\Phi)\nabla^2(\Phi - \Psi)$$

Dropping all nonlinear terms,

$$\alpha^2 R = \underbrace{6(\mathcal{H}' + \mathcal{H}^2)}_{\text{homogeneous}} - \underbrace{2\nabla^2\Psi + 4\nabla^2\Phi - 12(\mathcal{H}' + \mathcal{H}^2)\Psi - 6\Phi'' - 6\mathcal{H}(\Psi' + 3\Phi')}_{\text{linear correction}}$$

We work with the Einstein eqⁿ in the form $G^\mu{}_\nu = 8\pi G T^\mu{}_\nu$

First, let $\mu = \nu = 0$,

$$\begin{aligned} G^0{}_0 &= g^{0\lambda} G_{\lambda 0} = g^{00} \left[R_{00} - \frac{1}{2} g_{00} R \right] \\ &= -\alpha^2 (1 - 2\Psi) R_{00} - \frac{1}{2} R \quad (g^{0i} = 0 \text{ in Newtonian gauge, } g^{00} g_{00} = 1 + \mathcal{O}(2)) \\ &= -\alpha^2 (1 - 2\Psi) [-3\mathcal{H}' + \nabla^2\Psi + 3\mathcal{H}(\Phi' + \Psi') + 3\Phi''] \\ &\quad - \frac{1}{2} \alpha^2 [6(\mathcal{H}' + \mathcal{H}^2) - 2\nabla^2\Psi + 4\nabla^2\Phi - 12(\mathcal{H}' + \mathcal{H}^2)\Psi - 6\Phi'' - 6\mathcal{H}(\Psi' + 3\Phi')] \\ \Rightarrow \delta G^0{}_0 &= -\alpha^2 [6\mathcal{H}'\Psi + \nabla^2\Psi + 3\mathcal{H}(\Phi' + \Psi') + 3\Phi''] \\ &\quad + \alpha^2 [\nabla^2\Psi - 2\nabla^2\Phi + 6(\mathcal{H}' + \mathcal{H}^2)\Psi + 3\Phi'' + 3\mathcal{H}(\Psi' + 3\Phi')] \\ &= \alpha^2 [6\mathcal{H}\Phi' + 6\mathcal{H}^2\Psi - 2\nabla^2\Phi] \\ &= -2\alpha^2 [\nabla^2\Phi - 3\mathcal{H}(\Phi' + \mathcal{H}\Psi)] \end{aligned}$$

Then using $\delta G^\mu{}_\nu = 8\pi G \delta T^\mu{}_\nu$,

$$-2\alpha^2 [\nabla^2\Phi - 3\mathcal{H}(\Phi' + \mathcal{H}\Psi)] = 8\pi G (-\delta\rho)$$

$$\boxed{\nabla^2\Phi - 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G a^2 \delta\rho \quad (\delta\rho = \mathcal{L}_a \delta\rho_a)}$$

↳ The relativistic generalization of Poisson eqⁿ.

$$- \quad k \gg \mathcal{H} \Rightarrow |\nabla^2\Phi| \gg 3\mathcal{H}|\Phi' + \mathcal{H}\Psi| \xrightarrow[\text{to}]{\text{reduces}} \nabla^2\Phi \approx 4\pi G a^2 \delta\rho \quad (\text{Poisson eqⁿ in Newtonian limit})$$

Now inspect the case $\mu=i$ $\nu=j$:

$$\begin{aligned} G^i_j &= g^{ik} G_{kj} = g^{ik} \left[R_{kj} - \frac{1}{2} g_{kj} R \right] \\ &= \alpha^{-2} (1 + 2\Phi) \delta^{ik} R_{kj} - \frac{1}{2} \delta^i_j R \end{aligned}$$

The trace-free part gives $\partial_{\langle i} \partial_{j \rangle} (\Phi - \Psi) = 8\pi G a^2 \Pi_{ij} \Rightarrow \Psi \approx \Phi$

$\hookrightarrow$ Into Fourier space by the replacements $\partial_{\langle i} \partial_{j \rangle} \rightarrow -\kappa_{\langle i} \kappa_{j \rangle}$ and $\Pi_{ij} = -(\bar{\rho} + \bar{p}) \hat{k}_{\langle i} \hat{k}_{j \rangle} \Pi$ giving $\kappa^2 (\Phi - \Psi) = 8\pi G a^2 (\bar{\rho} + \bar{p}) \Pi$ ($\Pi = \mathcal{L}_a \Pi_a$)

Next, let $\mu=0$ $\nu=i$,

$$\begin{aligned} G^0_i &= g^{0\mu} G_{\mu i} = g^{00} R_{0i} \\ &= -\alpha^{-2} R_{0i} \\ &= -2\alpha^{-2} \partial_i (\Phi + \mathcal{H}\Psi) \end{aligned}$$

Using $\mathcal{T}^0_i = \partial_i q \Rightarrow \Phi' + \mathcal{H}\Psi = -4\pi G a^2 q$

Recall the Poisson eqⁿ, $\nabla^2 \Phi - 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G a^2 \delta\rho$

We can now write this as

$$\nabla^2 \Phi = 4\pi G a^2 \bar{\rho} \Delta$$

$\hookrightarrow$ Recall Δ is the comoving density contrast
 $\bar{\rho} \Delta = \delta\rho + \bar{\rho}'(v+B)$

Finally, we inspect the trace of the space-space Einstein eqⁿ

$$\Rightarrow \Phi'' + \frac{1}{3} \nabla^2 (\Psi - \Phi) + (2\mathcal{H}' + \mathcal{H}^2) \Psi + \mathcal{H}\Psi' + 2\mathcal{H}\Phi' = 4\pi G a^2 \delta\rho \quad (\delta\rho = \mathcal{L}_a \delta\rho_a)$$

If we assume $\Psi \approx \Phi$, this reduces to

$$\Phi'' + 3\mathcal{H}\Phi' + (2\mathcal{H}' + \mathcal{H}^2)\Phi = 4\pi G a^2 \delta\rho$$

$\hookrightarrow$ This becomes a closed eqⁿ of Φ if we use $\delta\rho = c_s^2 \delta\rho$ and the Poisson eqⁿ.